



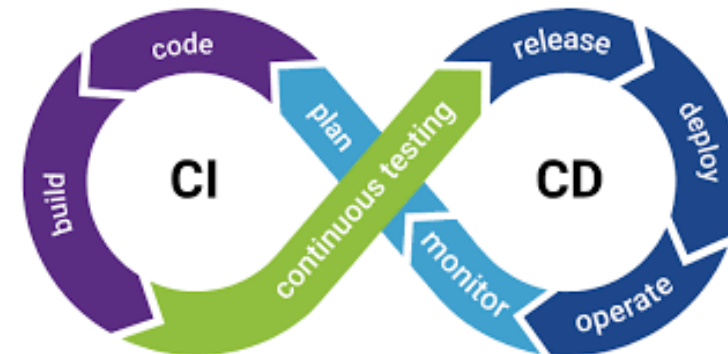
COM 682-Week 10

Cloud Native Development

- **Open-Source Licenses and Build Processes**
 - **CI/CD**

Agenda

- **Open-source license.**
 - Why chose an open-source license.
 - What are open-source licenses.
 - How to choose the “right” open-source license.
- **Build Processes.**
 - Continuous integration (CI) process.
 - Continuous deployment (CD) process.



Open-source Licenses

Overview

- An open-source license is a type of license for computer software and other products **that allows the source code, blueprint or design to be used, modified and/or shared under defined terms and conditions.**
- This allows end users and commercial companies to review and modify the source code, blueprint or design for their own customization, curiosity or troubleshooting needs.

Open-source Licenses

Overview



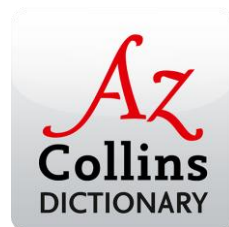
CODE = ART

From a legal perspective, all the code you write is considered a creative piece of work. It's the same as writing a book, writing a song, or producing a painting.

What is copyright?

“If someone has a copyright on a piece of writing or music, it is illegal to reproduce or perform it without their permission.”

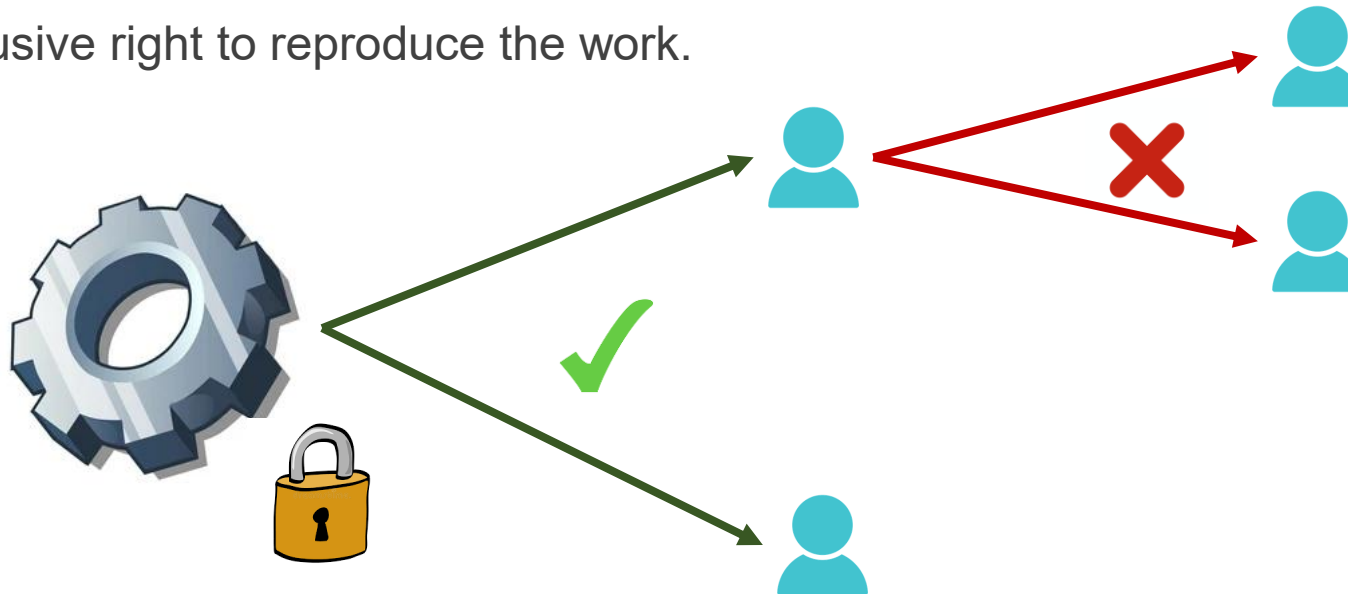
Collins dictionary



Open-source Licenses

What is Copyright?

- Copyright refers to **the legal right of the owner of intellectual property**.
- In simpler terms, **copyright is the right to copy**. This means that the original creators of products and anyone they give authorization to are the only ones with the exclusive right to reproduce the work.

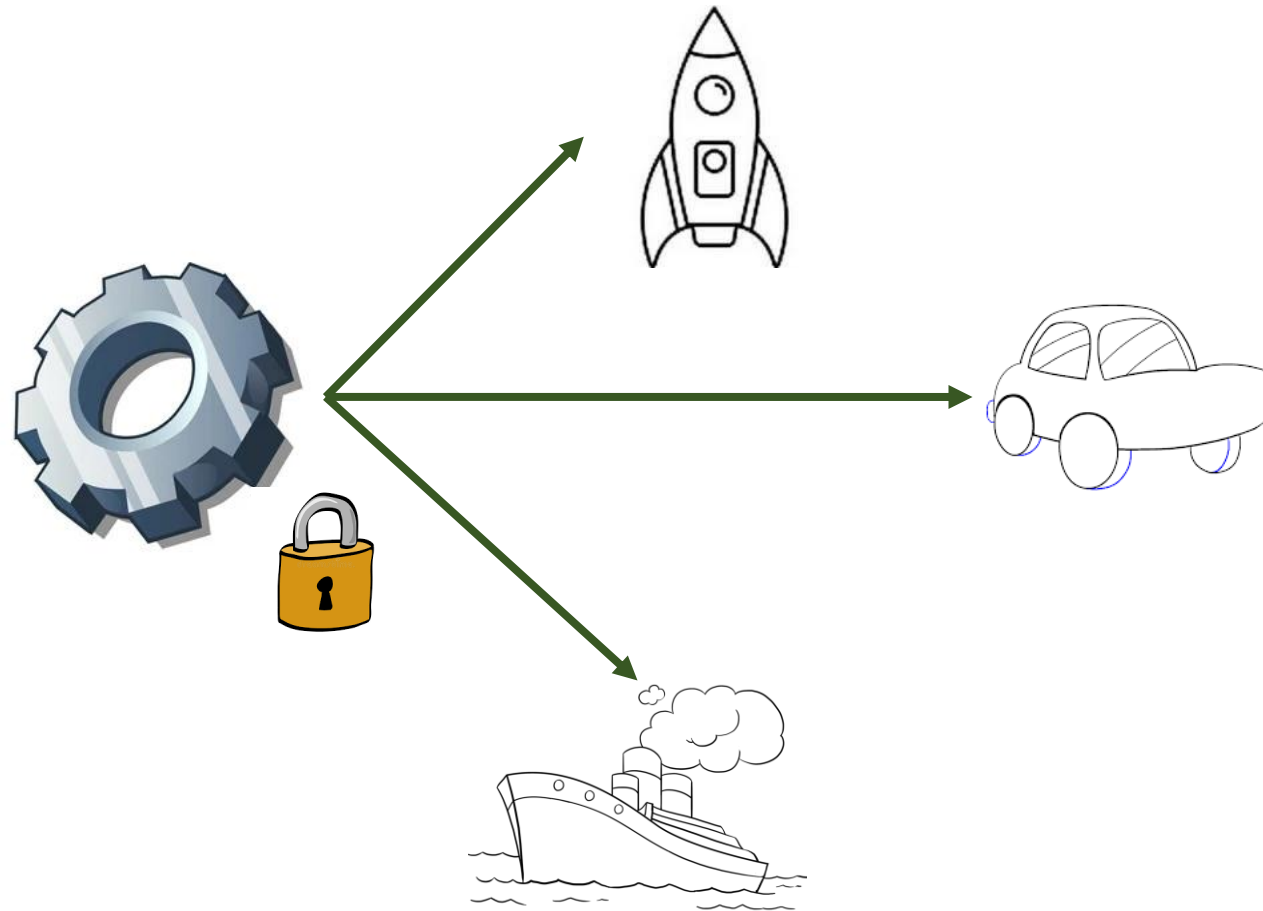


Copyright, protects your legal right as a creator to control who can Change/alter and distribute your software.

Open-source Licenses

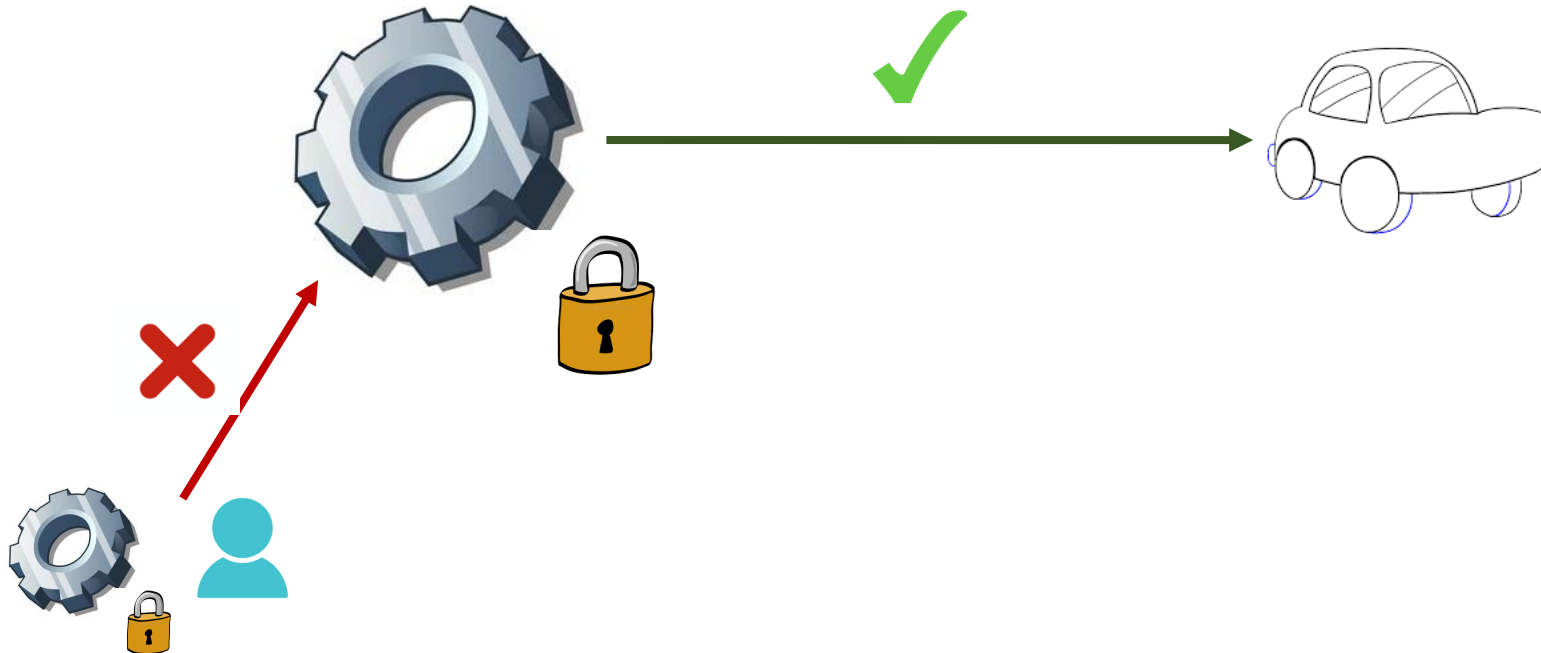
Copyright – Only you can define its use

- You can define copyright limitation to your code.



Open-source Licenses

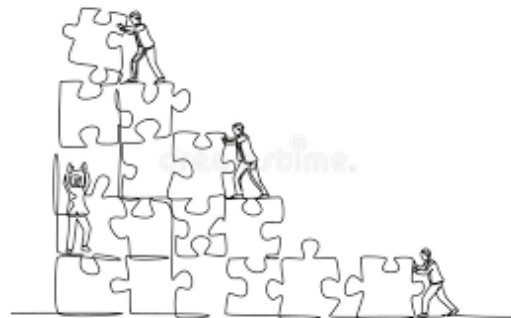
Copyright – Only you can create improvements to its use



Why do we write software?

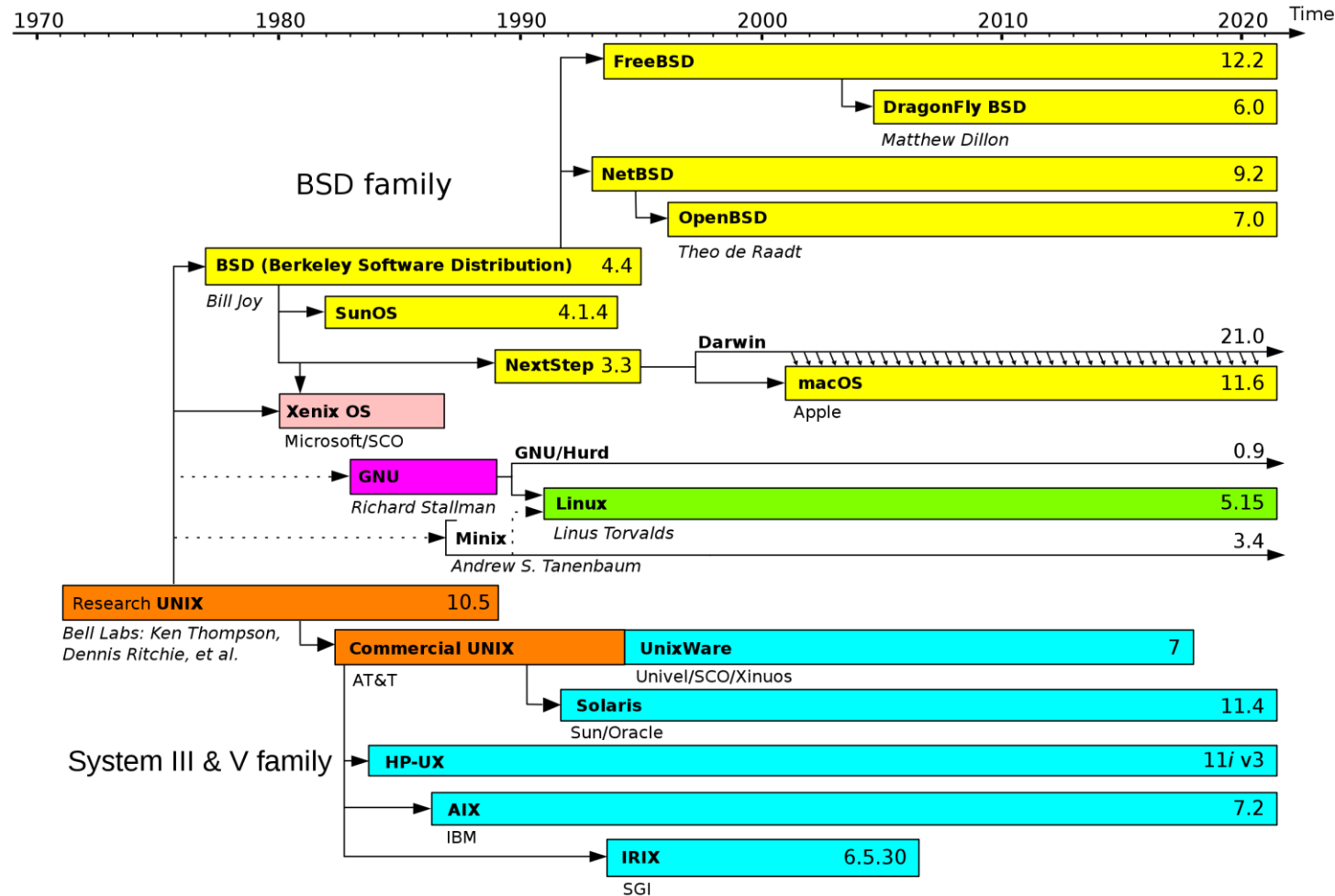
- To solve problems.

Opening Software to the World



Open Source != Free

Allow for Derivatives



Derivatives: An alteration or extension of a piece of software.

Open Source != Free

How to Make Money with Open Source?

Dual licensing

- Dual licensing allows companies to release commercial software (with a commercial license) that's derived from free OSS (open-source software) commonly distributed under the GNU General Public (GPL) license. Dual licensing can be implemented in a few ways. In the first scenario, a company releases identical products under a commercial license and under a free license like GPL. In the second scenario, the company releases different versions under different licenses.
- The most well-known example of successful dual licensing is **MySQL**. The company releases **MySQL Enterprise Edition**, MySQL Cluster CGE, and MySQL Standard Edition under commercial licenses while still offering two other products – **MySQL Classic Edition** and MySQL Community Edition – under the **GPL license**.
 - The commercial versions receive enhanced security features, backups, 24/7 support, and much more.

Open Source != Free

How to Make Money with Open Source?

Paid support

- Many OSS companies succeed by providing extra services.
- Such as technical support, certifications, and training. In fact, most professional open-source companies – including Red Hat, JBoss, and MySQL – have built their entire business by providing free solutions. They generate profits only from additional services.

Open-Source Initiative

License Criteria

- The license must be free to use.
- Must require access to source code.
- Must allow for the creation of altered works.
- Provide attribution/acknowledgment to the originator of work.
- No hidden conditions to provide royalties.
- Allow for the creation of derivatives.
- Disclaim authorship.

Popular Open-Source Licenses

- GPL (General Public License)
 - Linux
- LGPL (Lesser General Public License)
- Mozilla Public License 2.0
- MIT
- Apache 2.0

License Guidelines

- Use a license that has been approved by the OSI (Open-Source Initiative).
 - <https://opensource.org/licenses/>
- Read the license before you use it.
 - <https://tldrlegal.com/>
- Do not change your license once released.

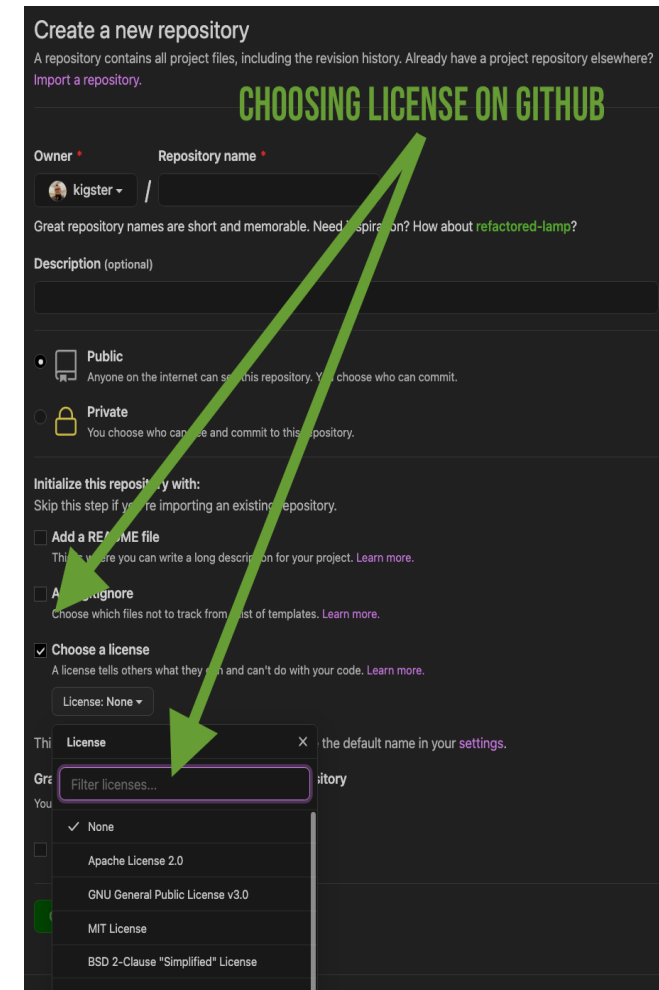
How to Apply your License?

- In the most common scenario, you will place a file named **LICENSE** at the root of your project sources.
- Many modern SaaS tools such as GitHub and GitLab will allow you to choose the license when the repository is first created. This is, perhaps, the single simplest way you can ensure your projects contain the right legal protections from the start.

Is a License Required Anywhere Else?

As a matter of fact — yes. In addition to placing the top-level LICENSE file, your project may also be subject to the specification requirements imposed by the *packaging and distribution format*.

```
# frozen_string_literal: true
#--
# Copyright (c) 2004–2021 David Heinemeier Hansson
#
# Permission is hereby granted, free of charge, to any person obtaining
# ... [stipped for brevity]
```



- Your open-source project can change the world.
- Allow your software to be freely used, modified, and shared, by using an open-source license.
- Select the right open-source license by asking the right questions.
 - <https://choosealicense.com/>

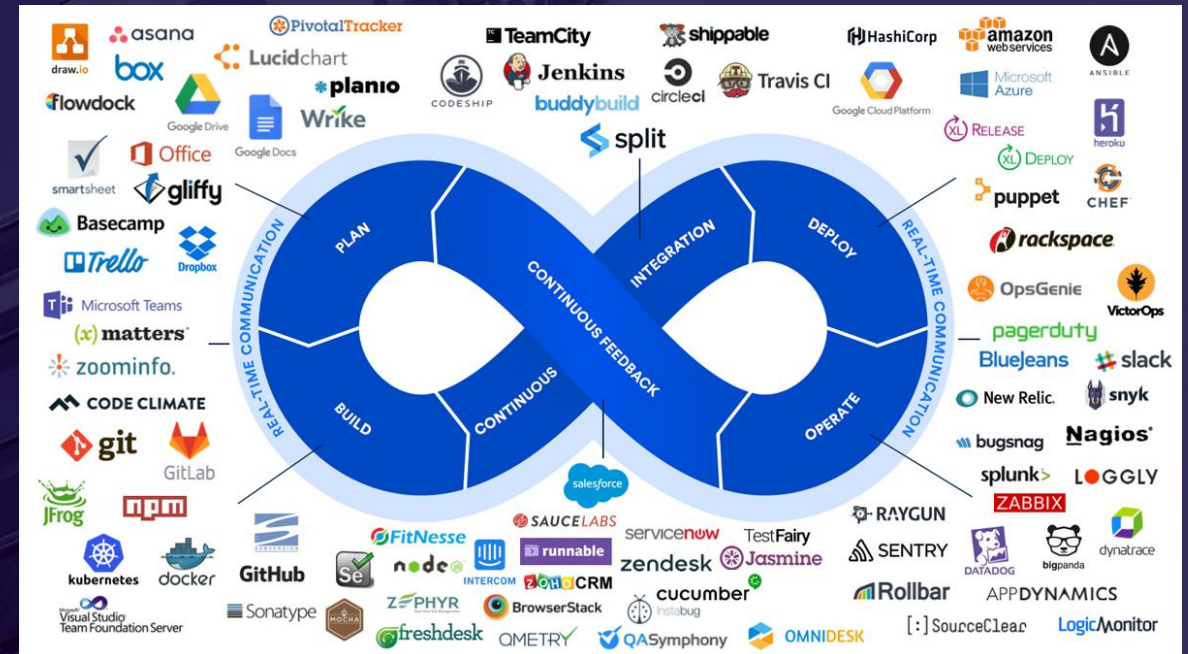
<https://opensource.org/faq#copyleft>



Open Source Initiative

Guaranteeing the 'our' in source...

Continuous Integration (CI) and Continuous Deployment (CD) Processes.



Definitions

- In software engineering, **continuous integration (CI)** is the practice of merging all developers' working copies to a shared mainline several times a day.
- **Continuous deployment (CD)** is a software engineering approach in which software functionalities are delivered frequently through automated deployments.
- **CD contrasts with continuous delivery**, a similar approach in which software functionalities are also frequently delivered and deemed to be potentially capable of being deployed **but are actually not deployed**.

Continuous Integration

- **Continuous Integration (CI)** is a development practice that requires developers to integrate code into a shared repository several times a day. Each check-in is then verified by an automated build, allowing teams to detect problems early.

The key details to note are that you need to run code integration multiple times a day, every day, and you need to run the automated verification of the integration.

What's the motivation for this? Well, in the development process, the earlier we surface errors, the better the end product will be.

- Continuous Integration **doesn't get rid of bugs**, but it does make them dramatically **easier to find and remove**.

Continuous Integration

1. Developers **check out code from the repository to work on it locally**. Ideally, they create a **new branch for the feature they want to implement**.
2. When their feature branch is ready, they **run tests locally in their development environments**.
3. Once all tests pass, they **push the commits** to the single source repository.
4. Whenever there are changes on the repository, a **CI server checks out the changes and performs a “build and test.”**
 - A build and test is when the CI server builds the entire system on the developer’s feature branch and runs all the unit and integration tests.
5. The **CI server notifies the team of the integration result**. There should generally be four outcomes: failed build, successful build, failed tests, and successful tests.
6. **If there’s a failure**, the team fixes the issue ASAP.
7. They **continue to develop** and repeat the steps 2–6 whenever there’s new code to be checked in to the repository.

Continuous Integration

Main principles

- Check in the code frequently.
- Automate the build and test portion.
- Always test the code locally before checking it in.
- Never merge any failed branches to the main branch.
- Return its status back to successful if you're the developer who causes the failed build or test.
- Make it your top priority to do so once the fail happens.

Continuous Deployment (CD)

- Within a development process, the deployment steps tend to repeat for each release. **Instead of performing the deployment manually for each release, CD considers the deployment as a step to be executed automatically.** Of course, ideally, this code has been built and tested successfully by the CI server too.

Continuous Deployment is closely related to Continuous Integration and refers to the release into production of software that passes the automated tests.

Essentially, it is the practice of releasing every good build to users.

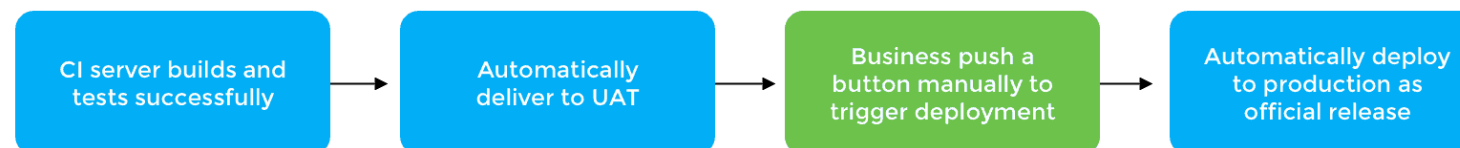
Difference Between CD and Continuous Delivery

- **Continuous deployment (CD)** is about automating the release of a good build to the production environment. On the other hand, **continuous delivery** is about ensuring that every good build is potentially ready for production release.

Continuous Deployment



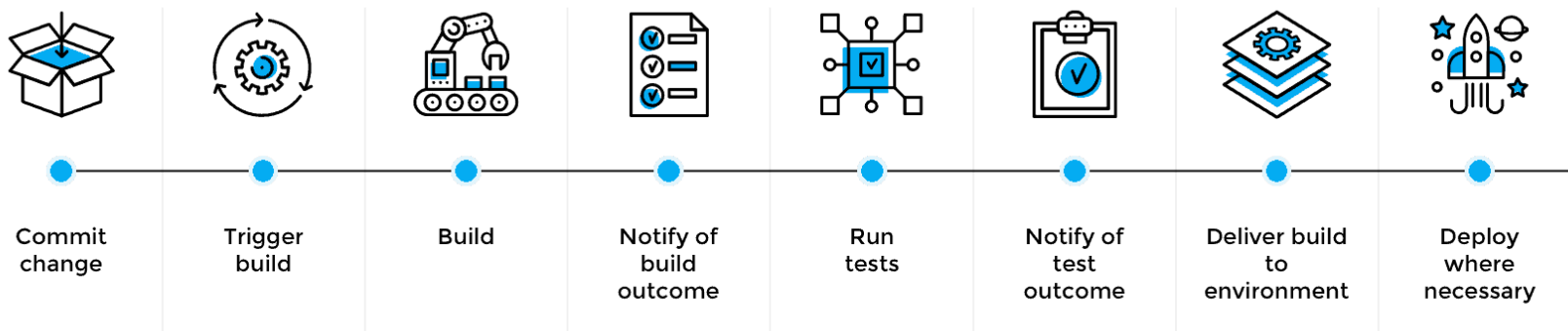
Continuous Delivery



CI/CD Pipeline

- The simplest version of the pipeline can best be described by this activity chain:

CI/CD Pipeline

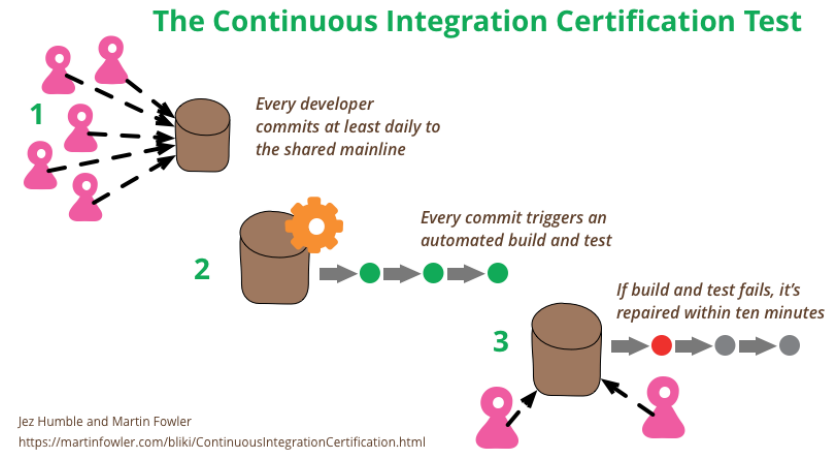


Why Does CI/CD Pipeline Matter?

- Reducing costs and complexity is a crucial benefit of employing a CI/CD pipeline.
- The CI/CD Pipeline Improves Reliability.

Combining the work of multiple developers is hard. Software systems are complex, and an apparently simple, self-contained change to a single file can easily have unintended consequences which compromise the correctness of the system. As a result, some teams have developers work isolated from each other on their own branches, both to keep trunk / main stable, and to prevent them treading on each other's toes.

— Jez Humble



- The CI/CD Pipeline Makes Your Team More Attractive to Developers



CI/CD in GitHub

What is GitHub Actions?

- Platform to automate developer **workflows**.
- CI/CD tool is one of many **workflows**.



What are those Workflows?



Platform for **open-source**
projects



- Publicly available to use and contribute to the project.
 - Which consequently generates **organizational tasks**.

What are those Workflows?

Organizational Tasks

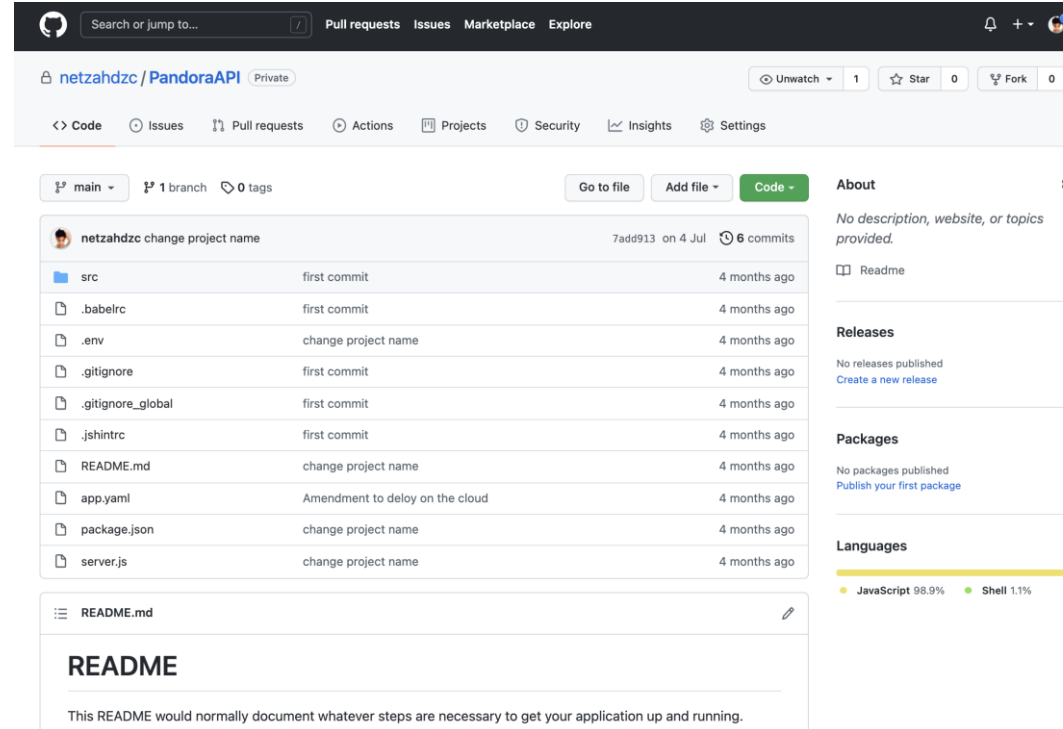


Contributors



Users

New Issue



- Is it minor / major?
- Is it reproducible?
- Assign to a contributor.

What are those Workflows?

Organizational Tasks



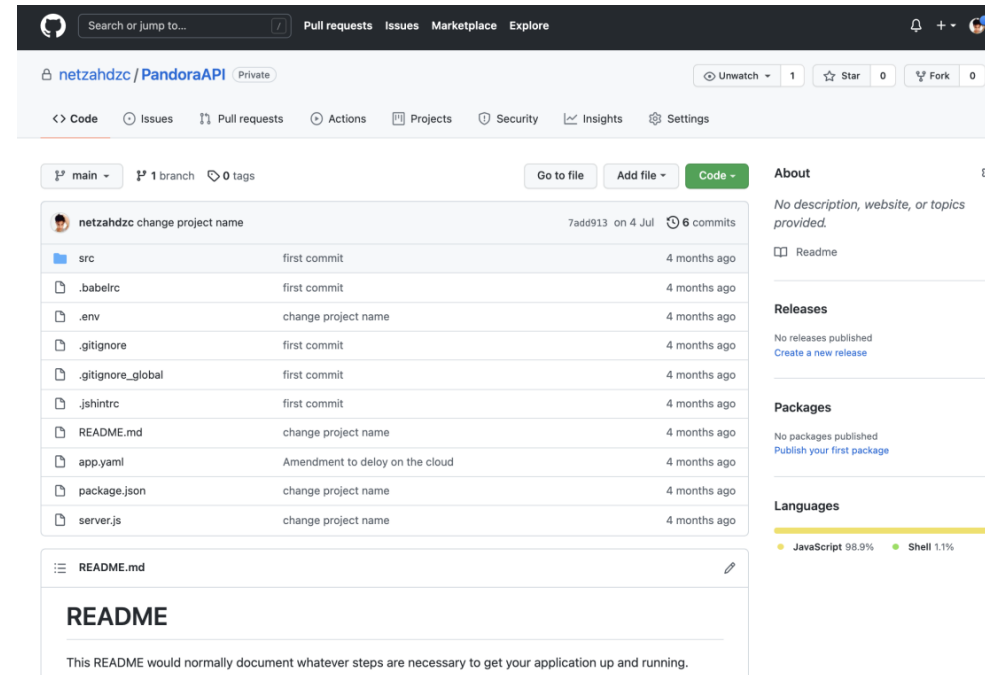
Contributors



New Issue



Users



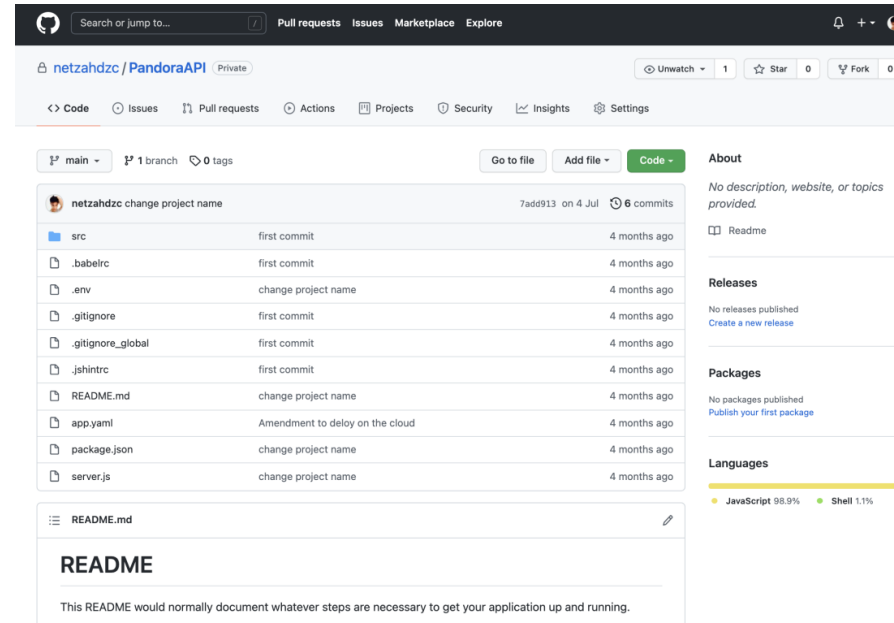
- Contributor sends a pull request.
- You review pull requests.
- Is the bug fixed?
- Merge to master branch.

What are those Workflows?

Organizational Tasks



Contributors



- Is it minor / major?
- Is it reproducible?
- Assign to a contributor.
- Contributor send a pull request.
- You review pull request.
- Is the bug fixed?
- Merge to master branch.
- Prepare release notes.
- Update version number.

These are three workflow examples, which complexity increase as more users/contributors you have; therefore more issues and pull requests.

How GitHub Actions Automate these Workflows?

- With GitHub actions, every time something happens (events) IN or TO your repository, automatic ACTIONS are executed in response.

1) Listen to Event

Pull request created

Pull request merged

Control joined

Issue created



Sort

Label

Assign it

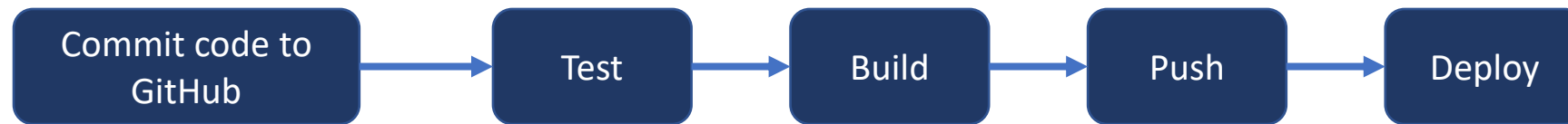
Reproduce

1 workflow consists of one or many actions

2) Trigger Workflow

CI/CD with GitHub Actions

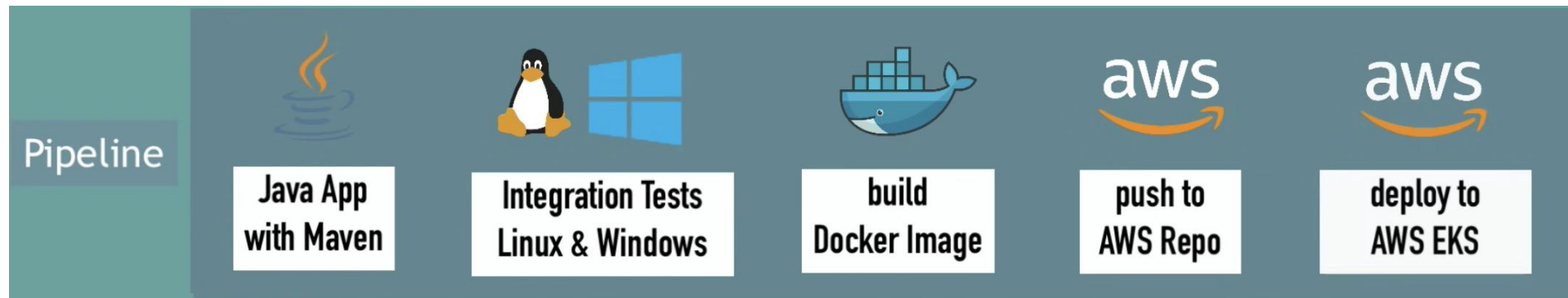
- Most common workflow for your repository.



CI/CD with GitHub Actions

Why to use GitHub?

- Use the **same tool** as the repository and build processes (i.e. CI/CD pipeline) instead of third-party integration.
- Setup the pipeline is easy.
- Too for developers / no need for an extra devOps fellow.
- Integration with other technologies is important.



- GitHub Actions facilitates the integration of all these tools:
 - ... with Node and Docker available.
 - ... with version you specify.
 - ... simply connect to target and deploy.



GitHub Example

Create Repo

 [netzahdzc](#) / [snake_game](#) Public

Unwatch 1 Star 0 Fork 0

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

[main](#) 1 branch 0 tags

[Go to file](#) [Add file](#) [Code](#)

 **netzahdzc** folder to tidy things cf786e0 4 hours ago 3 commits

 app	folder to tidy things	4 hours ago
 .gitignore	first commit	4 hours ago
 LICENSE	Initial commit	4 hours ago
 README.md	first commit	4 hours ago

 README.md 

Snake Game

This is a simple exampel to create a Readme file.

Installation

To run this game as localhost, the library [freegames](#) should be installed.

```
pip install freegames
```

About

No description, website, or topics provided.

 [Readme](#)

 [Apache-2.0 License](#)

Releases

No releases published
[Create a new release](#)

Packages

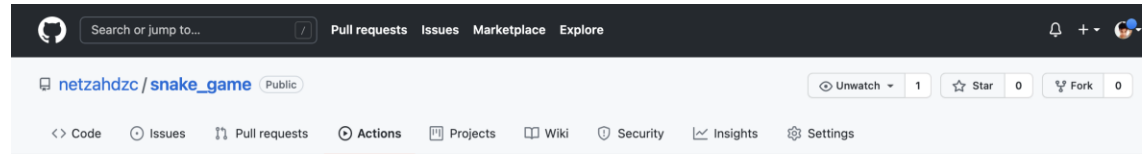
No packages published
[Publish your first package](#)

Languages

- Python 100.0%

GitHub Example

Workflow Templates



Get started with GitHub Actions

Build, test, and deploy your code. Make code reviews, branch management, and issue triaging work the way you want. Select a workflow template to get started.

Skip this and [set up a workflow yourself](#) →

Workflows made for your repository Suggested

Django

By GitHub Actions

Build and Test a Django Project

[Set up this workflow](#)

```
python -m pip install --upgrade pip
pip install -r requirements.txt
python manage.py test
```

actions/starter-workflows Python

Pylint

By GitHub Actions

Lint a Python application with pylint.

[Set up this workflow](#)

```
python -m pip install --upgrade pip
pip install pylint
pylint `ls -R | grep .py$ | xargs`
```

actions/starter-workflows Python

Python application

By GitHub Actions

Create and test a Python application.

[Set up this workflow](#)

```
python -m pip install --upgrade pip
pip install flake8 pytest
if [ -f requirements.txt ]; then pip install -r requirements.txt; fi
```

actions/starter-workflows Python

Python Package using Anaconda

By GitHub Actions

Create and test a Python package on multiple Python versions using Anaconda for package management.

[Set up this workflow](#)

```
# $CONDA is an environment variable pointing to the root of the
# miniconda directory
echo $CONDA/bin >> $GITHUB_PATH
conda env update --file environment.yml --name base
```

actions/starter-workflows Python

Deploy your code with these popular services

Deploy Node.js to Azure Web App

By Microsoft Azure

Build a Node.js project and deploy it to an Azure Web App.

[Set up this workflow](#)

actions/starter-workflows Deployment

Deploy to Alibaba Cloud ACK

By Alibaba Cloud

Deploy a container to Alibaba Cloud Container Service for Kubernetes (ACK).

[Set up this workflow](#)

actions/starter-workflows Deployment

Deploy to Amazon ECS

By Amazon Web Services

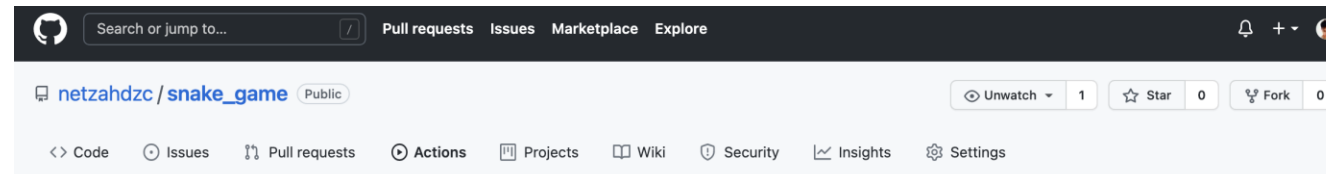
Deploy a container to an Amazon ECS service powered by AWS Fargate or Amazon EC2.

[Set up this workflow](#)

actions/starter-workflows Deployment

GitHub Example

Select Python Workflow



Get started with GitHub Actions

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conda env update --file environment.yml --name base
```

actions/starter-workflows Python

snake_game / .github / workflows python-app.yml in main

YAML

<https://www.cloudbees.com/blog/yaml-tutorial-everything-you-need-get-started>

```
<> Edit new file Preview Spaces 2 No wrap
1 # This workflow will install Python dependencies, run tests and lint with a single version of Python
2 # For more information see: https://help.github.com/actions/language-and-framework-guides/using-python-with-gith
3
4 name: Python application
5
6 on:
7   push:
8     branches: [ main ]
9   pull_request:
10    branches: [ main ]
11
12 jobs:
13   build:
14
15     runs-on: ubuntu-latest
16
17     steps:
18     - uses: actions/checkout@v2
19     - name: Set up Python 3.9
20       uses: actions/setup-python@v2
21       with:
22         python-version: 3.9
23     - name: Install dependencies
24       run: |
25         python -m pip install --upgrade pip
26         pip install flake8 pytest
27         if [ -f requirements.txt ]; then pip install -r requirements.txt; fi
28     - name: Lint with flake8
29       run: |
30         # stop the build if there are Python syntax errors or undefined names
31         flake8 . --count --select=E9,F63,F7,F82 --show-source --statistics
32         # exit-zero treats all errors as warnings. The GitHub editor is 127 chars wide
33         flake8 . --count --exit-zero --max-complexity=10 --max-line-length=127 --statistics
34     - name: Test with pytest
35       run: |
36         pytest
37
```

Name of the workflow

Events

Workflow

Search or jump to... Pull requests Issues Marketplace Explore

netzahdzc / snake_game Public

<> Code Issues Pull requests Actions Projects Wiki Security Insights

snake_game / .github / workflows / python-app.yml in main

<> Edit new file Preview Spaces 2 No map

```

1 # This workflow will install Python dependencies, run tests and lint with a single version of Python
2 # For more information see: https://help.github.com/actions/language-and-framework-guides/using-python-with-github-actions
3
4 name: Python application
5
6 on:
7   push:
8     branches: [ main ]
9   pull_request:
10    branches: [ main ]
11
12 jobs:
13   build:
14
15     runs-on: ubuntu-latest
16
17     steps:
18     - uses: actions/checkout@v2
19     - name: Set up Python 3.9
20       uses: actions/setup-python@v2
21       with:
22         python-version: 3.9
23     - name: Install dependencies
24       run: |
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35       run: |
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37

```

Search or jump to... Pull requests Issues Marketplace Explore

GitHub Actions

Automate your GitHub workflows

https://github.com/features/actions (Verified)

Overview Repositories 43 Packages 18 Projects

Pinned

- starter-workflows Accelerating new GitHub Actions workflows
- toolkit The GitHub Toolkit for developing GitHub Actions
- setup-node Set up your GitHub Actions workflow with a specific version of node.js
- javascript-action Create a JavaScript Action with tests, linting, workflow, publishing, and versioning
- typescript-action Create a TypeScript Action with tests, linting, workflow, publishing, and versioning
- labeler An action for automatically labelling pull requests

Repositories

1 result for repositories matching che sorted by last updated

checkout Public Action for checking out a repo

Type Language Sort

View all repositories

Step 1 of the workflow

Notes:

- “uses” attribute when calling a repository.
- “run” attribute when running commands.

netzahdzc / snake_game Public

Unwatch 1 Star 0 Fork 0

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

snake_game / .github / workflows / python-app.yml

Cancel changes Start commit

Edit new file

Preview

Spaces 2 No wrap

```
1 # This workflow will install Python dependencies, run tests and lint with a single version of Python
2 # For more information see: https://help.github.com/actions/language-and-framework-guides/using-python-with-github-actions
3
4 name: Python application
5
6 on:
7   push:
8     branches: [ main ]
9   pull_request:
10    branches: [ main ]
11
12 jobs:
13   build:
14
15     runs-on: ubuntu-latest
16
17     steps:
18     - uses: actions/checkout@v2
19     - name: Set up Python 3.9
20       uses: actions/setup-python@v2
21       with:
22         python-version: 3.9
23     - name: Install dependencies
24       run: |
25         python -m pip install --upgrade pip
26         pip install flake8 pytest
27         if [ -f requirements.txt ]; then pip install -r requirements.txt; fi
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34     - name: Test with pytest
35       run: |
36         pytest
37
```

Commit new file

Create python-app.yml

Add an optional extended description...

- ☐ Commit directly to the `main` branch.
- ☒ Create a new branch for this commit and start a pull request. [Learn more about pull requests.](#)

netzahdzc-patch-1

Propose new file

- By actions 
- Setup a Go environment and add it to the PATH
-  **Setup .NET Core SDK**  360
- By actions 
- Used to build and publish .NET source. Set up a specific version of the .NET and authentication to private NuGet repository
-  **First interaction**  152
- By actions 
- Greet new contributors when they create their first issue or open their first pull request

Featured categories

Code quality

Monitoring

[Pull requests](#) [Issues](#) [Marketplace](#) [Explore](#)[netzahdzc / snake_game](#) PublicUnwatch 1 Star 0 Fork 0[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

Open a pull request

The change you just made was written to a new branch named `netzahdzc-patch-1`. Create a pull request below to propose these changes.

base: main compare: netzahdzc-patch-1 ✓ Able to merge. These branches can be automatically merged.

Write PreviewH B I ≡ <> 🔗 ⋮ ⋮ ☑ @ 📎 ↶

Leave a comment

Attach files by dragging & dropping, selecting or pasting them.

Create pull request

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

Reviewers

No reviews

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Linked issues

Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

[GitHub Community Guidelines](#)

1 commit

1 file changed

0 comments

1 contributor

Create python-app.yml #1

Edit

<> Code

Open

 netzahdzc wants to merge 1 commit into `main` from `netzahdzc-patch-1`

Conversation 0

Commits 1

Checks 0

Files changed 1

+36 -0

 netzahdzc commented now

Owner

No description provided.

  Create python-app.yml

Verified 49849d0

Add more commits by pushing to the `netzahdzc-patch-1` branch on `netzahdzc/snake_game`.

Some checks haven't completed yet

1 in progress check

Python application / build (pull_request)

In progress — This check has started...

Details

This branch has no conflicts with the base branch

Merging can be performed automatically.

Merge pull request

You can also [open this in GitHub Desktop](#) or view [command line instructions](#).

Write

Preview

H B I  <>       

Leave a comment

Attach files by dragging & dropping, selecting or pasting them.

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Linked issues

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Unsubscribe

You're receiving notifications because you're watching this repository.



Create python-app.yml #1

Edit

<> Code

Open

 netzahdzc wants to merge 1 commit into `main` from `netzahdzc-patch-1`

Conversation 0

Commits 1

Checks 1

Files changed 1

+36 -0

 netzahdzc commented 3 minutes ago

Owner

No description provided.

 Create python-app.yml

Verified 49849d0

Add more commits by pushing to the `netzahdzc-patch-1` branch on `netzahdzc/snake_game`.

 **All checks have failed** [Hide all checks](#)
1 failing check

  Python application / build (pull_request) Failing after 10s — build [Details](#)

 **This branch has no conflicts with the base branch**
Merging can be performed automatically.

Merge pull request

You can also [open this in GitHub Desktop](#) or view [command line instructions](#).

 Write Preview

H B I         

Leave a comment

Attach files by dragging & dropping, selecting or pasting them.

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Linked issues

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Unsubscribe

You're receiving notifications because you're watching this repository.

1 participant





 [netzahdzc / snake_game](#) Public

Unwatch 1 Star 0 Fork 0

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

Workflows

New workflow

All workflows

 Python application

All workflows

Showing runs from all workflows

Filter workflow runs

11 workflow runs			Event ▾	Status ▾	Branch ▾	Actor ▾
✓	Update snake.py	Python application #13: Commit 387fad9 pushed by netzahdzc	main		21 seconds ago 19s	...
✓	Update snake.py	Python application #12: Commit 311316e pushed by netzahdzc	main		1 minute ago 22s	...
✓	Update python-app.yml	Python application #11: Commit bc2a1a6 pushed by netzahdzc	main		2 minutes ago 22s	...
✗	Update python-app.yml	Python application #10: Commit 4de26dc pushed by netzahdzc	main		5 minutes ago 26s	...
✗	Update python-app.yml	Python application #9: Commit b0978b1 pushed by netzahdzc	main		8 minutes ago 21s	...
✗	Update snake.py	Python application #8: Commit f4fd529 pushed by netzahdzc	main		11 minutes ago 22s	...
✗	Update python-app.yml	Python application #7: Commit ceae473 pushed by netzahdzc	main		13 minutes ago 22s	...
✓	Update snake.py				19 minutes ago	

 main ▾ [snake_game](#) / [.github](#) / [workflows](#) / [python-app.yml](#)

[View runs](#) [Go to file](#) [...](#)

 **netzahdzc** Update python-app.yml ✓ Latest commit bc2a1a6 6 minutes ago [History](#)

 1 contributor

33 lines (28 sloc) | 1.07 KB [Raw](#) [Blame](#) 

```
1 # This workflow will install Python dependencies, run tests and lint with a single version of Python
2 # For more information see: https://help.github.com/actions/language-and-framework-guides/using-python-with-github-actions
3
4 name: Python application
5
6 on:
7   push:
8     branches: [ main ]
9   pull_request:
10    branches: [ main ]
11
12 jobs:
13   build:
14
15     runs-on: ubuntu-latest
16
17     steps:
18     - uses: actions/checkout@v2
19     - name: Set up Python 3.9
20       uses: actions/setup-python@v2
21       with:
22         python-version: 3.9
23     - name: Install dependencies
24       run: |
25         python -m pip install --upgrade pip
26         pip install flake8
27         if [ -f requirements.txt ]; then pip install -r requirements.txt; fi
28     - name: Lint with flake8
29       run: |
30         # stop the build if there are Python syntax errors or undefined names
31         flake8 . --count --select=E9,F63,F7,F82 --show-source --statistics
32         # exit-zero treats all errors as warnings. The GitHub editor is 127 chars wide
33         flake8 . --count --exit-zero --max-complexity=10 --max-line-length=127 --statistics
```

Where does this code run?

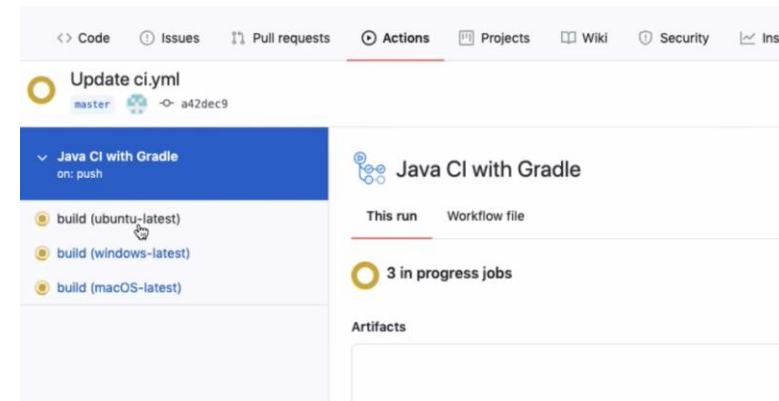
- Managed by GitHub
 - Each job/workflow runs in a single server; **workflow runs in parallel by default.**

To override the default parallel execution, type:

```
publish:  
  needs: build
```

To run the code in different OS.

```
runs-on: ${{matrix.os}}  
strategy:  
  matrix:  
    os: [ubuntu-latest, windows-latest, macOS-  
latest]
```

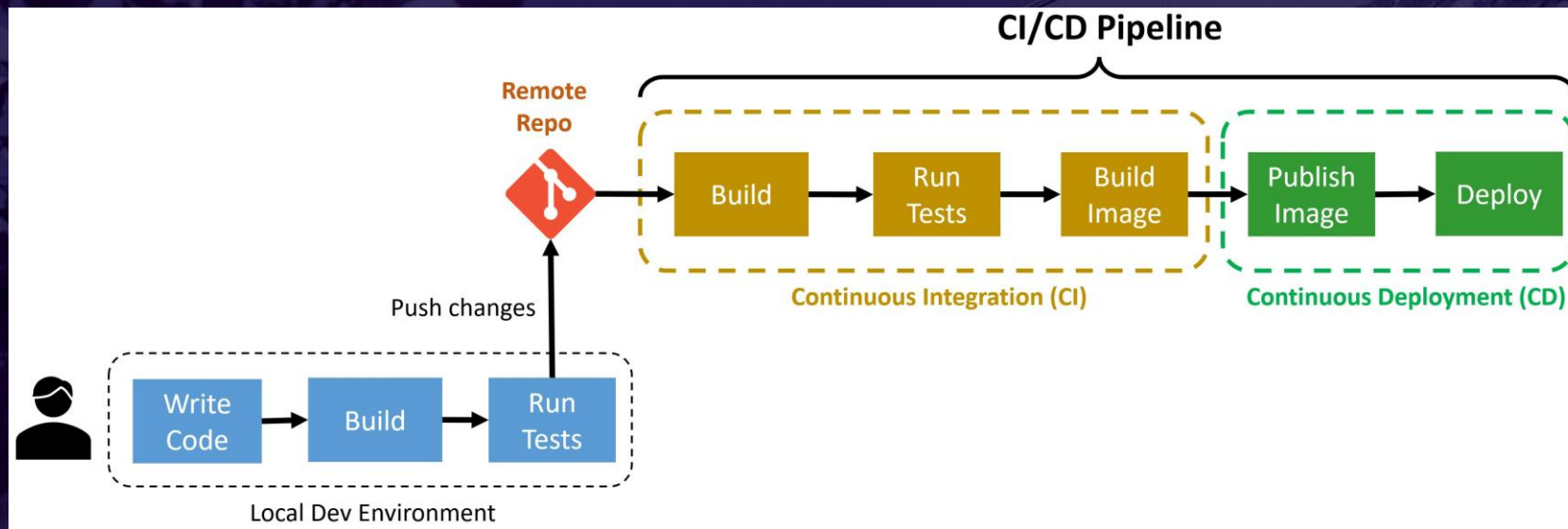


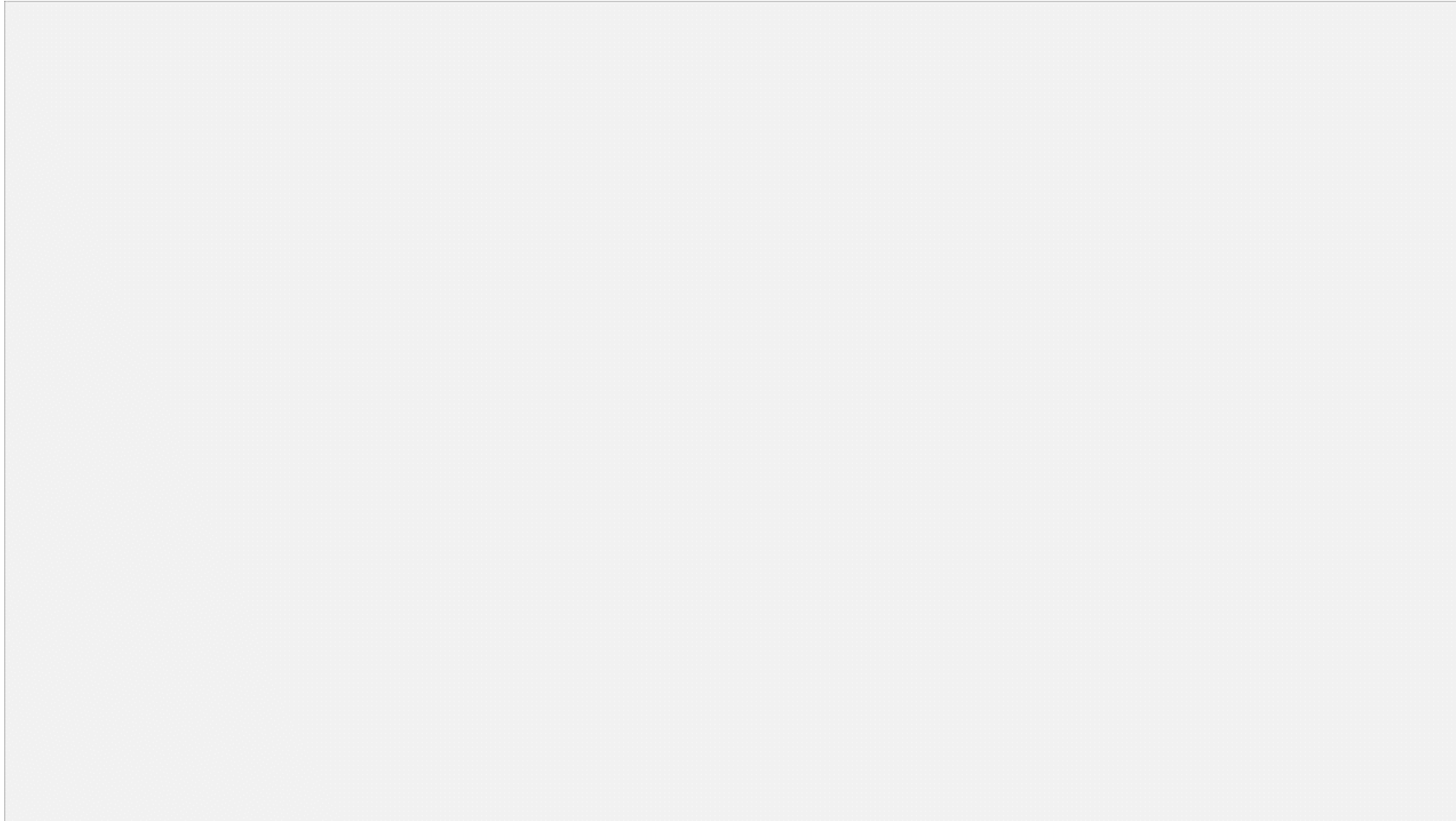


References

- <https://docs.github.com/en>
- <https://docs.github.com/en/actions>

CI/CD pipeline in Azure





Azure Pipelines



Azure Pipelines

- The Azure CI/CD pipeline simplifies continuous integration and continuous delivery (CI/CD) in the application development process. You can start from the source stage with existing code on GitHub or on-premise containers.
- The Azure Repos can maintain a central repository, and the Azure pipelines maintain build and release pipelines for the given project. The Azure DevOps CI/CD process is a crucial process with all the required dev services.
- Apart from continuous integration and continuous deployment with Azure DevOps, these pipelines are used to construct build-deploy-test workflows used mainly in continuous testing (CT). This tests the changes in a fast and scalable routine.

- DevOps Starter makes it easy to get started on Azure using either GitHub actions or Azure DevOps. It helps you launch your favourite app on the Azure service of your choice in just a few quick steps from the Azure portal.
- DevOps Starter sets up everything you need for developing, deploying, and monitoring your application. You can use the DevOps Starter dashboard to monitor code commits, builds, and deployments, all from a single view in the Azure portal.
- You can use DevOps Starter to:
 - Quickly deploy your application to Azure
 - Automate the setup of a CI/CD workflow or pipeline
 - View and understand how to properly set up a CI/CD workflow or pipeline
 - Further customize the release pipelines based on your specific scenarios

Creating an ASP .NET CI/CD Project using an Azure Pipeline

- Azure DevOps project simplifies the setup of CI/CD pipelines in Azure. You can go with the existing code or create a sample application as a new Azure DevOps project.

Step-1: Prerequisites

Make sure you have the following items:

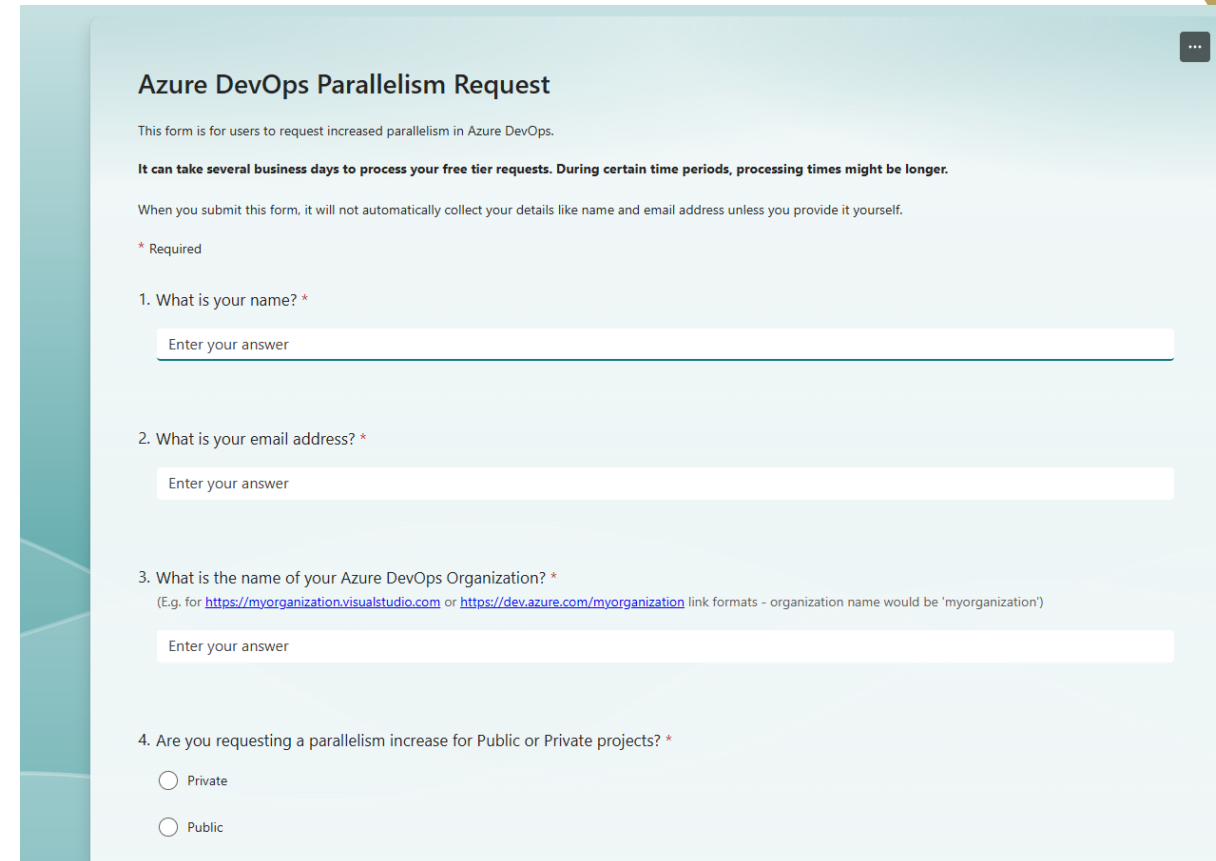
A GitHub account where you can create a repository. [Create one for free.](#)

An ability to run pipelines on Microsoft-hosted agents. If you haven't done this before then

please fill out the following form <https://aka.ms/azpipelines-parallelism-request>

It will take 2-3 days to get the response and approve your grant

An Azure DevOps organization using this link: [Create one for free.](#)



Azure DevOps Parallelism Request

This form is for users to request increased parallelism in Azure DevOps.

It can take several business days to process your free tier requests. During certain time periods, processing times might be longer.

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

1. What is your name? *
2. What is your email address? *
3. What is the name of your Azure DevOps Organization? *
(E.g. for <https://myorganization.visualstudio.com> or <https://dev.azure.com/myorganization> link formats - organization name would be 'myorganization')
4. Are you requesting a parallelism increase for Public or Private projects? *
☐ Private
☐ Public

Version

Azure DevOps Services

Filter by title

Azure Pipelines

Get started

What is Azure Pipelines?

Sign up for Azure Pipelines

Create your first pipeline

Customize your pipeline

Pipeline basics

Pipeline structure & logic

Agents & pools

Build & Deploy

Test

Monitor & Troubleshoot

Learn / Azure / Azure DevOps / Azure Pipelines /

Sign up for Azure Pipelines

Article • 03/26/2024 • 16 contributors

In this article

Sign up with a Microsoft account

Sign up with a GitHub account

Create a project

Invite team members - optional

Show 3 more

Azure DevOps Services

Sign up for an Azure DevOps organization and Azure Pipelines to begin managing CI/CD to deploy your code with high-performance pipelines.

Additional resources

Training

Learning path

Get started with Azure DevOps learning path - Training

In this learning path, find out how Azure DevOps enables transparency and cooperation with continuous delivery and deployment for your development lifecycle.

Certification

Microsoft Certified: DevOps Engineer Expert - Certifications

This certification measures your ability to accomplish the following technical tasks: Design and implement processes and communications, design and implement a source control...

Documentation

What is Azure Pipelines? - Azure Pipelines

Learn how to automatically build, test, and deploy your code

Click on ‘Free Account’

On next page, click on ‘Start Free’ under Azure pipelines:

azure.microsoft.com/en-us/pricing/details/devops/azure-devops-services/

Microsoft Azure Explore Products Solutions Pricing Partners Resources Learn Support Contact Sales Get started with Azure Sign in All Microsoft Search

Pricing for Azure DevOps

Azure DevOps Services Azure DevOps Server

Prices are estimates only and are not intended as actual price quotes. Actual pricing may vary depending on the type of agreement entered with Microsoft, date of purchase, and the currency exchange rate. Prices are calculated based on US dollars and converted using London closing spot rates that are published in the last business day prior to the last business day of the previous month and if the last business day prior to the end of the month falls on a bank holiday in major markets, the rate setting day is generally the day immediately preceding the last business day. This rate applies to all transactions during the upcoming month. Sign in to the [Azure pricing calculator](#) to view pricing based on your current program offer with Microsoft. Contact us [Azure sales specialists](#) for more information on pricing or to request a price quote. See [Azure specific add-on pricing](#) without Azure pricing.

Country United States - Dollar (\$) USD

INDIVIDUAL SERVICES

Azure Pipelines

1 Free Microsoft-hosted CI/CD
1 Free Self-Hosted CI/CD

Start free

1 Microsoft-hosted job with 1,800 minutes per month for CI/CD and 1 self-hosted job with unlimited minutes per month
\$40 per extra Microsoft-hosted CI/CD parallel job and \$15 per extra self-hosted CI/CD parallel job with unlimited minutes

Microsoft-hosted

Azure Artifacts

2 GB free, then starting at \$2 per GB

Start free

Industry-leading NuGet Server
Support for Maven, npm, and Python packages
Upstream sources to help protect open-source dependencies
Integrated with Azure Pipelines
Sophisticated access controls

Free 2 GB free

GitHub Advanced Security for Azure DevOps

\$49 per contributor per month

Activate now

Full suite of security tools native to Azure DevOps
Secret scanning
Code scanning
Dependency scanning

Total contributors

USER LICENSES

Basic Plan

First 5 users free, then \$8 per user per month

Start free

Azure Pipelines: Includes the free offer from INDIVIDUAL SERVICES
Azure Boards: Work item tracking and Kanban boards
Azure Repos: Unlimited private Git repos
Azure Artifacts: 2 GB free per organization

First 5 users free. For additional users, see table

Basic - Test Plans

\$52 per user per month

30 day free trial

Includes all Basic plan features
Test planning, tracking & execution
Browser-based tests with annotation
Rich-client test execution
User acceptance testing
Centralized reporting

\$52 per user per month

You will be asked for the project name and visibility

organization policies.' At the bottom is a blue button with a plus icon and the text 'Create project'." data-bbox="170 66 495 481"/>

Create a project to get started

Project name *

COM682 CI-CD Pipeline

Visibility

☒ Public
Anyone on the internet can view the project. Certain features like TFVC are not supported.

☐ Private
Only people you give access to will be able to view this project.

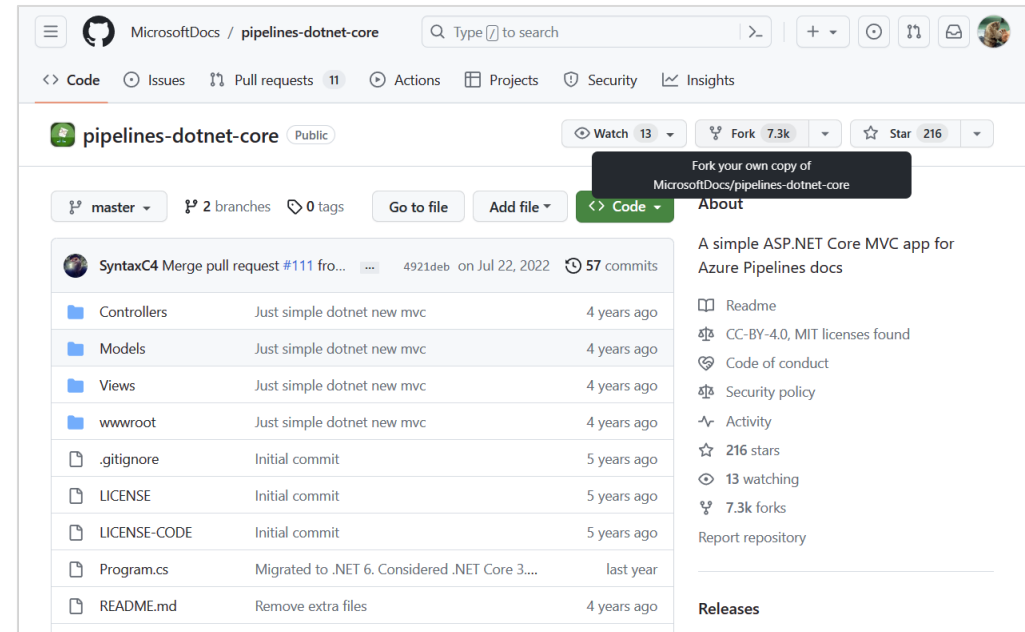
Public projects are disabled for your organization. You can turn on public visibility with [organization policies](#).

+ Create project

Click on Create Project

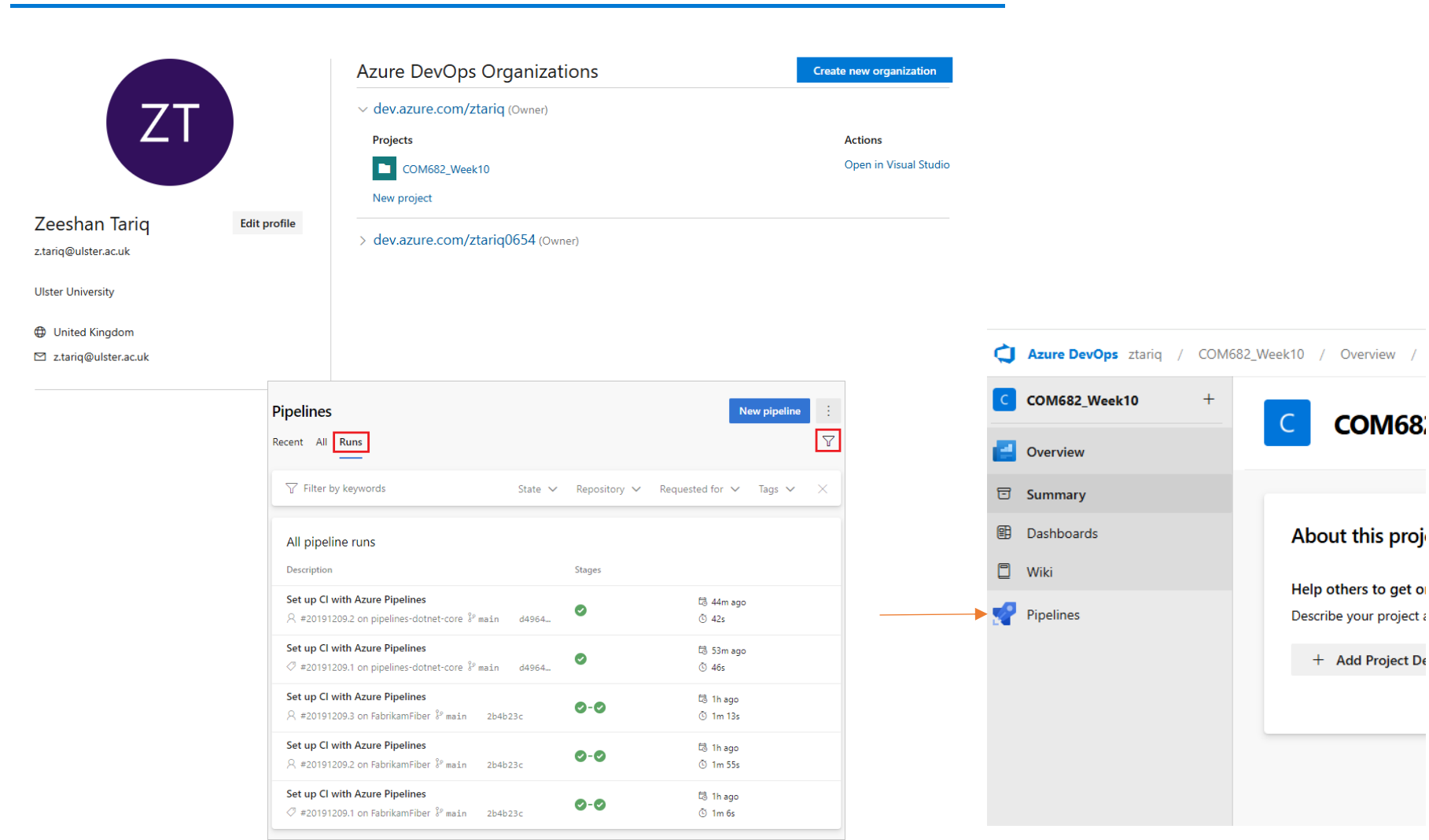
Once this step is completed you can view your organization and project at <https://aex.dev.azure.com/me?mkt=en-US>

- **Step-2: Cloning a target repository into your GitHub Account**



Step-3: Setting up a DevOps project on Azure

Access your Azure DevOps Organisation you created previously via <https://aex.dev.azure.com/me?mkt=en-US> .In this case, we have set up the ztariq organisation and project name is COM682_week10 for this practical.



The screenshot displays the Azure DevOps user interface. On the left, the user profile for Zeeshan Tariq is shown, including their email (z.tariq@ulster.ac.uk) and affiliation with Ulster University. The main area is titled 'Azure DevOps Organizations' and lists two organizations: 'dev.azure.com/ztariq' (Owner) and 'dev.azure.com/ztariq0654' (Owner). Under the first organization, a project named 'COM682_Week10' is listed with an 'Open in Visual Studio' link. Below this, the 'Pipelines' section is visible, showing a list of pipeline runs. The 'Runs' tab is selected, and a table of pipeline runs is displayed. An orange arrow points from the 'Runs' tab in the Pipelines section to the 'Pipelines' option in the left-hand navigation menu.

Azure DevOps Organizations

Create new organization

dev.azure.com/ztariq (Owner)

Projects

COM682_Week10

Open in Visual Studio

New project

dev.azure.com/ztariq0654 (Owner)

Pipelines

New pipeline

Recent All **Runs**

Filter by keywords

State Repository Requested for Tags

All pipeline runs

Description	Stages	Time
Set up CI with Azure Pipelines #20191209.2 on pipelines-dotnet-core main d4964...	✓	44m ago 42s
Set up CI with Azure Pipelines #20191209.1 on pipelines-dotnet-core main d4964...	✓	53m ago 46s
Set up CI with Azure Pipelines #20191209.3 on FabrikamFiber main 2b4b23c	✓-✓	1h ago 1m 13s
Set up CI with Azure Pipelines #20191209.2 on FabrikamFiber main 2b4b23c	✓-✓	1h ago 1m 55s
Set up CI with Azure Pipelines #20191209.1 on FabrikamFiber main 2b4b23c	✓-✓	1h ago 1m 6s

COM682_Week10

Overview

Summary

Dashboards

Wiki

Pipelines

About this project

Help others to get on

Describe your project

+ Add Project De



Questions

